



PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064



A MACHINE LEARNING-BASED GROUNDWATER LEVEL PREDICTOR FOR SUSTAINABLE WATER MANAGEMENT FOR KARNATAKA

A PROJECT REPORT

Submitted by

CHIRAG K S - 20221CSD0134

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SHREYAS V - 20221CSD0117

Under the guidance of,

Dr. Yamanappa

BACHELOR OF TECHNOLOGY

IN

**COMPUTER SCIENCE AND ENGINEERING
(DATA SCIENCE)**

PRESIDENCY UNIVERSITY

BENGALURU

DECEMBER 2025



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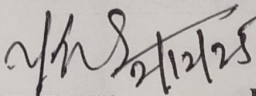
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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

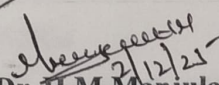
Certified that this report "A Machine Learning-Based Groundwater Level Predictor for Sustainable Water Management" is a bonafide work of "CHIRAG K S(20221CSD0134), ARJUN ATHREY(20221CSD0133), SHREYAS V(20221CSD0117)", who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE ENGINEERING, DATA SCIENCE during 2025-26.


Dr. Yamanappa

Project Guide

PSCS

Presidency University

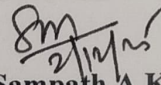

Dr. H M Manjula

Program Project

Coordinator

PSCS

Presidency University


Dr. Sampath A K

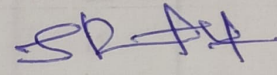
Dr. Geetha A

School Project

Coordinators

PSCS

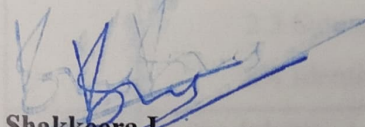
Presidency University


Dr. Pravinthraja S

Head of the Department

PSCS(CSD)

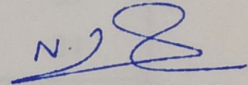
Presidency University


Dr. Shakkeera L

Associate Dean

PSCS

Presidency University

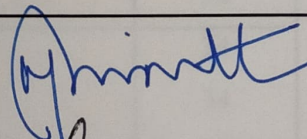
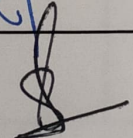

Dr. Duraipandian N

Dean

PSCS & PSIS

Presidency University

Name and Signature of the Examiners

1)	Dr.K Marimuthu	Professor Presidency School of Computer Science and Engineering	
2)	Ms.Sharon M	Assistant Professor Presidency School of Computer Science and Engineering	

PRESIDENCY UNIVERSITY

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

We the students of final year B.Tech in COMPUTER SCIENCE ENGINEERING, CYBER SECURITY at Presidency University, Bengaluru, named CHIRAG K S, SHREYAS V, ARJUN ATHREY hereby declare that the project work titled "A Machine Learning-Based Groundwater Level Predictor for Sustainable Water Management" has been independently carried out by us and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE ENGINEERING (Data Science) during the academic year of 2025-26. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

CHIRAG K S

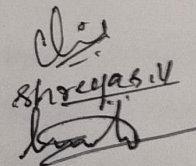
USN: 20221CSD0134

SHREYAS V

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ARJUN ATHREY

USN: 20221CSD0133



PLACE: BENGALURU

DATE: 1-December 2025

ABSTRACT

This project proposal is a documentation of designing a machine learning-based Groundwater Level Predictor of Karnataka with an emphasis on sustainable management of the water resources. This is aimed at reducing the acute stress on groundwater resources through increased demand and climate change through the provision of predictive means by which proactive stewardship can be achieved. It is expected that a sound model will be created that will predict the groundwater level with accuracy. It will include integrating a combination of multiple sources into an integrated spatio-temporal dataset, deploying and testing the most cutting-edge machine learning models, which will be mainly Random Forest, and subjecting them to a severe evaluation process, and deriving practical conclusions about the key factors regarding the causes of the dynamics of groundwater levels. The deliverables will include a proven predictive model, an extensive dataset, and a detailed technical report, which will help the governmental agencies to allocate data in water, prepare against droughts, and assess the policies, and thus improve the administrative management of the Karnataka water future.

The ultimate goal would be to develop a plausible model of predicting the monthly groundwater levels at the taluk scale using a heterogeneous spatio-temporal dataset that is gathered together. It will implement and test the state-of-the-art machine learning models, whereby the Random Forest has been identified as one of the most promising models and put them through rigorous performance testing. Besides predictive accuracy, the research will also determine and measure the main drivers affecting the alterations in the groundwater levels and thus produce actionable information in water resource management.

The deliverables will include a certified prediction model, a complete dataset, and a technical report of the results. The outputs will enable the government agencies to make evidence-based decisions on the water allocations, drought preparedness and policy making thus translating into long term sustainability and viability of water resources in Karnataka.

Checklist

- Students to print checklist separately.
- Confirming the checklist is only for ensuring completeness, correctness and quality of report. Use the checklist to confirm all aspects are fulfilled.
- Confirming the checklist does not assure full marks. In addition to report, marks are awarded for other aspects of the project viz. implementation and presentation.
- **Do not include this checklist in the report.**

A. Confirm **Formatting** – line spacing, paragraph justification, font size.

B. Confirm **Text** – details, spellings, grammar

C. Confirm **Figures and Tables** - in-text citation, description

D. Confirm **Figures and Tables** – updated in List of Figures and Tables

E. Confirm **Citation** – all sources referred are cited

F. Confirm **References** –all sources referred are updated in Reference section

Faculty to mark 'X' in columns not confirming

Sl. No.	Description	A	B	C	D	E	F
	Declaration	✓	✓	X	X	X	X
	Acknowledgement	✓	✓	X	X	X	X
	Abstract	✓	✓	X	X	X	X
	Table of content	✓	✓	✓	✓	X	X
	– confirm Section, sub-sections from all chapters included.	✓	✓	✓	✓	X	X
	– confirm Page no. of Section, sub-sections updated	✓	✓	✓	✓	X	X
	– confirm Page no. are aligned to right	✓	✓	X	X	X	X
	List of Figures	✓	✓	✓	✓	X	X
	– confirm Caption of Figures from all chapters included.	✓	✓	✓	X	X	X
	– confirm Page no. of figures updated	✓	✓	✓	✓	X	✓
	– confirm Page no. are aligned to right	✓	✓	✓	X	X	X
	– confirm Figures are numbered correctly	✓	✓	✓	✓	X	X
	List of Tables	✓	✓	✓	✓	X	X

	- confirm Caption of Tables from all chapters included.	✓	✓	✓	✗	✗	✗
	- confirm Page no. of tables updated	✓	✓	✓	✓	✗	✗
	- confirm Page no. are aligned to right	✓	✓	✓	✗	✗	✗
	- confirm Tables are numbered correctly	✓	✓	✓	✓	✗	✗
	Abbreviations	✓	✓	✗	✗	✗	✗
	- confirm Abbreviations used in all chapters included	✓	✓	✗	✗	✗	✗
	- confirm Abbreviations are in alphabetic order.	✓	✓	✗	✗	✗	✗
	- confirm Abbreviations appear only once in list and are not repeated	✓	✓	✗	✗	✗	✗

- A. Confirm **Formatting** – line spacing, paragraph justification, font size.
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Sl. No.	Description	A	B	C	D	E	F
1.	Introduction	✓	✓	✓	✓	✓	✗
	1.1 Background	✓	✓	✗	✗	✓	✗
	1.2 Statistics of project	✓	✓	✗	✗	✓	✗
	1.3 Prior existing technologies	✓	✓	✗	✗	✓	✗
	1.4 Proposed approach	✓	✓	✗	✗	✓	✗
	1.5 SDGs	✓	✓	✗	✓	✓	✗
2.	1.6 Overview of project report	✓	✓	✗	✗	✗	✗
	Literature review	✓	✓	✗	✗	✓	✗
3.	Methodology	✓	✓	✗	✓	✓	✗

	3.1 Overview of Methodologies	✓	✓	✓	✓	✓	✓	✓
	3.2 Comparative Analysis of Methodologies	✓	✓	✓	✓	✓	✓	✓
	3.3 Selected Methodology	✓	✓	✓	✓	✓	✓	✓
	3.4 Customized V-Model for the Project	✓	✓	✓	✓	✓	✓	✓
4.	Project Management	✓	✓	✓	✓	✓	✓	✓
	4.1 Project Timeline	✓	✓	✓	✓	✓	✓	✓
	4.2 Risk Analysis	✓	✓	✓	✓	✓	✓	✓
	4.3 Project Budget	✓	✓	✓	✓	✓	✓	✓
5.	Analysis and Design	✓	✓	✓	✓	✓	✓	✓
	5.1 Requirements	✓	✓	✓	✓	✓	✓	✓
	5.2 Block Diagram	✓	✓	✓	✓	✓	✓	✓
	5.3 System Flow Chart	✓	✓	✓	✓	✓	✓	✓
	5.4 Algorithms and Tools	✓	✓	✓	✓	✓	✓	✓
	5.5 Unit Design	✓	✓	✓	✓	✓	✓	✓
6.	5.6 Functional View	✓	✓	✓	✓	✓	✓	✓
	Software and Simulation	✓	✓	✓	✓	✓	✓	✓
	6.1 Software Development Tools	✓	✓	✓	✓	✓	✓	✓
	6.2 Pseudo-code for Algorithms	✓	✓	✓	✓	✓	✓	✓
	6.3 Simulation	✓	✓	✓	✓	✓	✓	✓

A. Confirm **Formatting** – line spacing, paragraph justification, font size.

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SL No.	Description	A	B	C	D	E	F
7.	Evaluation and Results	✓	✓	✗	✓	✗	✗

I confirm that I have uploaded all the required documents for the final project submission as per the given instructions. The same has been verified and approved by my guide.

CHIRAG KS

Team Leader Name and Signature



Dr. Yamanappa

Guide Name and Signature



31/12/2025

	- confirm Caption of Tables from all chapters included.	✓	✓	✓	✗	✗	✗
	- confirm Page no. of tables updated	✓	✓	✓	✓	✗	✗
	- confirm Page no. are aligned to right	✓	✓	✓	✗	✗	✗
	- confirm Tables are numbered correctly	✓	✓	✓	✓	✗	✗
	Abbreviations	✓	✓	✗	✗	✗	✗
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